

```

37.     # Generate signals
38.     data['Signal'] = 0
39.     data.loc[data['SMA_Short'] > data['SMA_Long'], 'Signal'] = 1
40.
41.     # Run backtest
42.     backtest_result = backtest_strategy(data)
43.
44.     # Evaluate results
45.     metrics = evaluate_backtest(backtest_result)
46.
47.     # Store results
48.     result = {
49.         'short_ma': short_ma,
50.         'long_ma': long_ma,
51.         **metrics
52.     }
53.     results.append(result)
54.
55.     # Convert results to DataFrame
56.     results_df = pd.DataFrame(results)
57.
58.     # Sort by Sharpe Ratio (or another preferred metric)
59.     results_df = results_df.sort_values('Sharpe Ratio', ascending=False)
60.
61.     return results_df
62.
63. Example usage:
64. # Define parameter grid
65. param_grid = {
66.     'short_ma': [5, 10, 15, 20, 25, 30],
67.     'long_ma': [50, 75, 100, 125, 150, 200]
68. }
69.
70. # Perform grid search
71. results = grid_search_optimization('AAPL', '2018-01-01', '2023-01-01', param_grid)
72.
73. # Display top 5 parameter combinations
74. print(results.head(5))
75.

```